Exp 16: EXPERIMENT TO UNDERSTAND EDA-QUALITATICE AND QUANTITATIVE ANALYSIS.

AIM: Understand the importance of EDA-Quantitative and Qualitative analysis.

SAMPLE CODE:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

%matplotlib inline

# Load the dataset
tips = sns.load_dataset('tips')

# Display first five rows
print(tips.head())

# 1. Distribution plot with KDE
sns.displot(tips.total_bill, kde=True)
plt.show()

# 2. Distribution plot without KDE
sns.displot(tips.total_bill, kde=False)
plt.show()
```

```
# 3. Joint plot
sns.jointplot(x=tips.tip, y=tips.total_bill)
plt.show()

# 4. Joint plot with regression line
sns.jointplot(x=tips.tip, y=tips.total_bill, kind='reg')
plt.show()

# 5. Joint plot with hexagonal bins
sns.jointplot(x=tips.tip, y=tips.total_bill, kind='hex')
plt.show()

# 6. Pair plot
sns.pairplot(tips)
plt.show()

# 7. Pair plot grouped by time
sns.pairplot(tips, hue='time')
plt.show()

# 8. Pair plot grouped by day
sns.pairplot(tips, hue='day')
plt.show()

# 9. Display time value counts
print(tips.time.value_counts())
```

```
# 10. Correlation heatmap
sns.heatmap(
   tips.corr(numeric_only=True),
   annot=True
)
plt.show()

# 11. Box plot of total bill
sns.boxplot(y=tips.total_bill)
plt.show()

# 12. Box plot of tip
sns.boxplot(y=tips.tip)
plt.show()

# 13. Count plot of day
sns.countplot(x='day', data=tips)
plt.show()

# 14. Count plot of sex
sns.countplot(y='sex', data=tips)
plt.show()

# 15. Pie chart of sex
tips.sex.value_counts().plot(kind='pie')
plt.show()

# 16. Bar chart of sex
```

```
tips.sex.value_counts().plot(kind='bar')
plt.show()

# 17. Count plot of days for Dinner
sns.countplot(
   x='day',
   data=tips[tips.time == 'Dinner']
)
plt.show()
```

OUTPUT:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

tips=sns.load_dataset('tips')

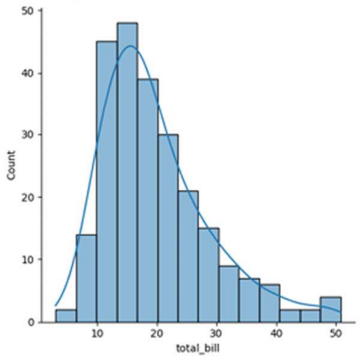
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

+ Code + Text

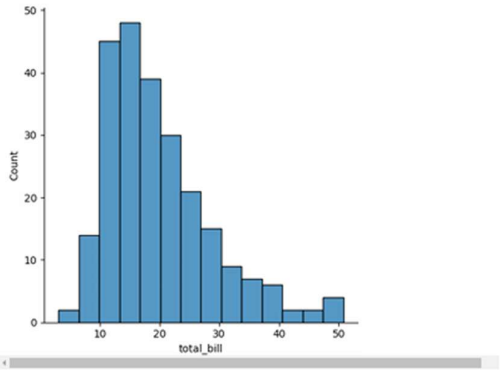
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```



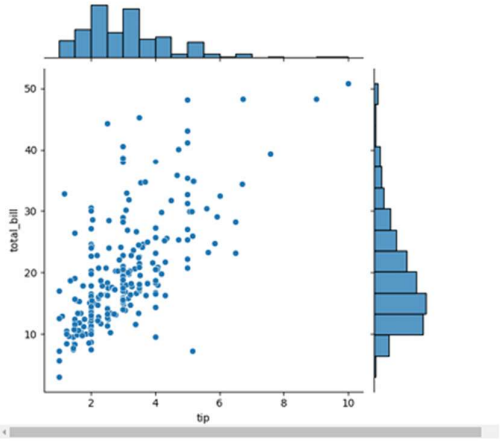

```
sns.displot(tips.total_bill,kde=False)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb0b0af580>
```



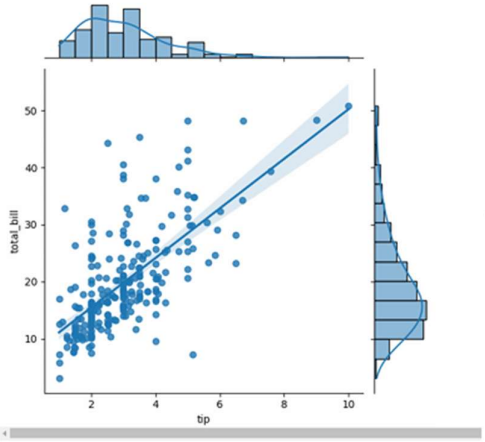

```
sns.jointplot(x=tips.tip,y=tips.total_bill)
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc96c0>
```



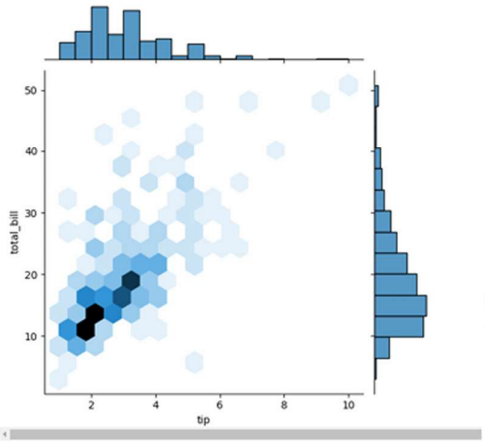

```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc9cf0>
```



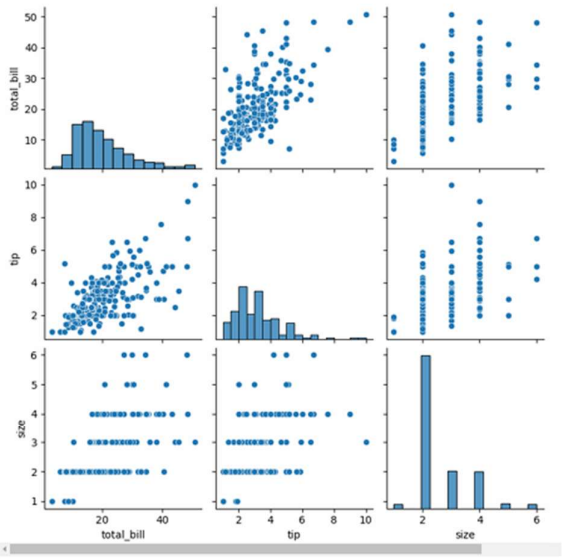

```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
```

```
<seaborn.axisgrid.JointGrid at 0x79bb088f4730>
```




```
sns.pairplot(tips)
```

```
<seaborn.axisgrid.PairGrid at 0x79bb06fc3d30>
```

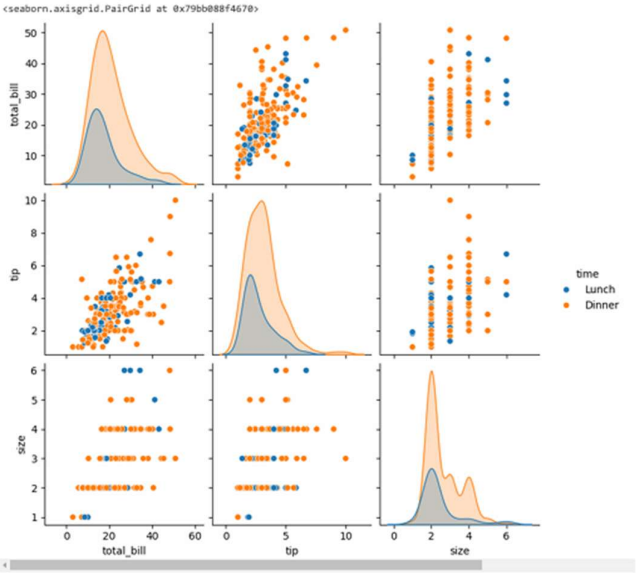



```
tips.time.value_counts()
```

time	count
Dinner	176
Lunch	68

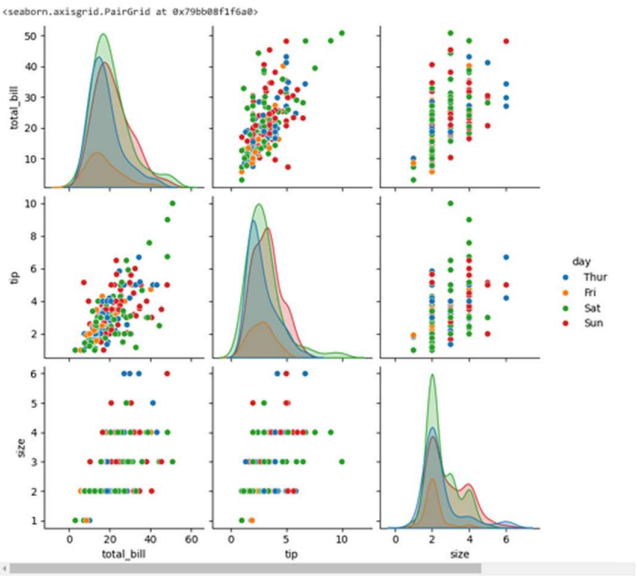
```
sns.pairplot(tips,hue='time')
```

```
<seaborn.axisgrid.PairGrid at 0x79bb088f4670>
```



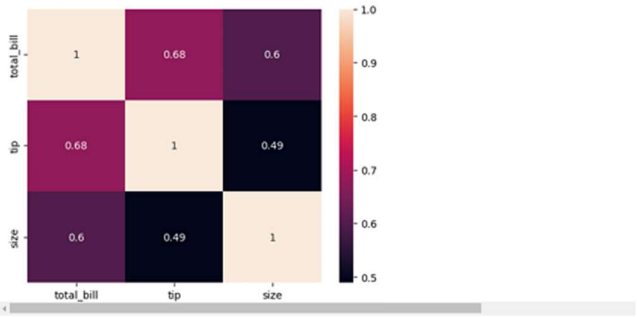

```
sns.pairplot(tips,hue='day')
```

```
<seaborn.axisgrid.PairGrid at 0x79bb08f1f6a0>
```



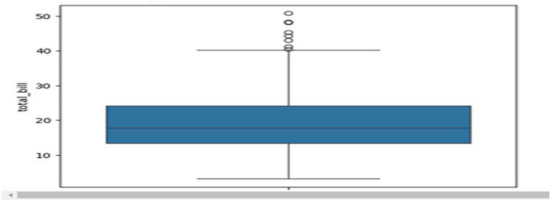

```
sns.heatmap(tips.corr(numeric_only=True),annot=True)
```

```
<Axes: >
```



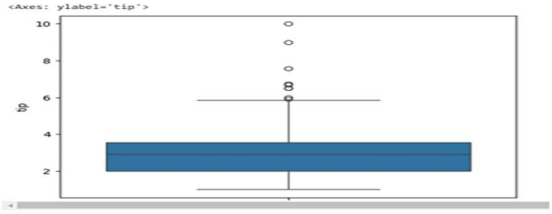

```
sns.boxplot(tips.total_bill)
```

```
<Axes: ylabel='total_bill'>
```



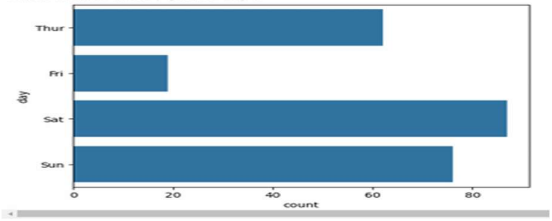

```
sns.boxplot(tips.tip)
```

```
<Axes: ylabel='tip'>
```



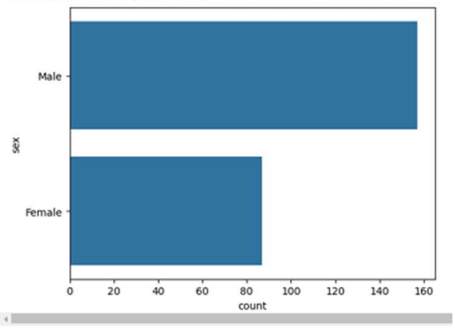

```
sns.countplot(tips.day)
```

```
<Axes: xlabel='count', ylabel='day'>
```

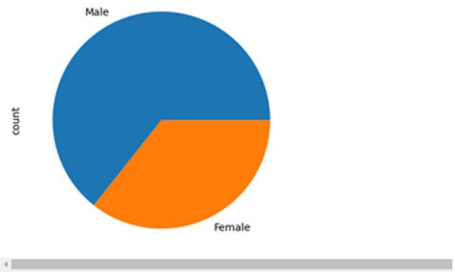



```
sns.countplot(tips.sex)
```

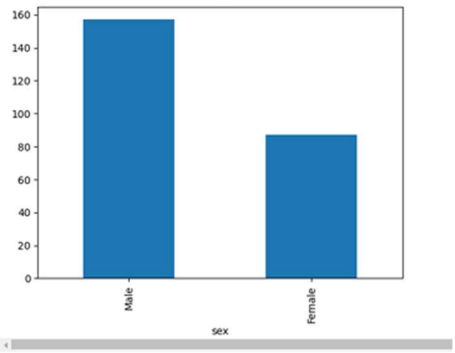
```
<Axes: xlabel='count', ylabel='sex'>
```




```
tips.sex.value_counts().plot(kind='pie')
<Axes: ylabel='count'>
```




```
tips.sex.value_counts().plot(kind='bar')
<Axes: xlabel='sex'>
```




```
sns.countplot(tips[tips.time=='Dinner']['day'])
<Axes: xlabel='count', ylabel='day'>
```



RESULT: Thus, experiment to understand EDA- qualitative and quantitative analysis is conducted.